E-Commerce Price Tracker – Insights Report

1. Project Overview

This project analyzes laptop prices collected from an e-commerce platform (Amazon) using web scraping. The data was cleaned and analyzed using Python and visualized through a dashboard built in Microsoft Power BI Desktop. The goal is to understand price distribution, product ratings, and market trends.

2. Dataset Summary

Metric	Value
Total Products	82
Platform	Amazon
Data Source	Web Scraping
Fields	Product Name, Price, Rating, Review Count, Platform, Date

3. Key Insights

Insight 1 – Product Price Distribution

Most laptops fall into the **mid-range price category**.

Observation:

1. Budget laptops (< ₹40,000) are fewer
2. Most products are between **₹40,000 – ₹60,000**

Note: Mid-range laptops dominate the online market.

Insight 2 – Rating vs Price Relationship

High-priced laptops generally have **better ratings**.

Observation:

- Premium laptops often have **ratings above 4**
- Budget laptops sometimes have lower ratings

Note: High-priced often indicates these laptops have better features.

Insight 3 – Market Price Range

The majority of laptops are priced between:

₹40,000 – ₹60,000

Note: This is the **most competitive pricing segment** in the laptop market.

Insight 4 – Top Expensive Products

The dashboard shows the **most expensive laptops available on the platform**.

Observation:
Premium gaming or high-performance laptops appear in this category.

Note: These products target **high-end consumers**.

Insight 5 – Product Ratings

Average rating of laptops is approximately: 3.5 – 4.5

Note: Most laptops available online have **good customer satisfaction**.

4. Business Value

This analysis helps:

- Track **market pricing trends**
- Identify **popular price segments**
- Understand **customer rating behavior**
- Support **competitive pricing strategies**

Businesses can use this data to:

- Price products competitively
- Identify high-demand price ranges
- Monitor competitor pricing.

5. Dashboard Insights

The dashboard created in Power BI provides:

- Total number of products tracked
- Average product price
- Average customer rating
- Price distribution across categories
- Top expensive laptops
- Price vs rating relationship

This helps users quickly analyze the **e-commerce laptop market**.

6. Future Improvements

The project can be improved by:

- Scraping multiple platforms (Amazon, Flipkart)
- Tracking prices daily or weekly
- Adding brand analysis
- Detecting price drops automatically
- Creating automated dashboards

7. Conclusion

The E-Commerce Price Tracker successfully collects and analyzes laptop pricing data from an online marketplace. The analysis shows that mid-range laptops dominate the market and higher-priced laptops tend to receive better ratings. The dashboard helps visualize these trends and supports data-driven decision making.